

A shared measure of the health of intelligence

Critical research overview and candidate measurement interface

Prepared for Albert Jan van Hoek | 7 October 2026

Status: Critical narrative review and instrument-development proposal. The proposed instrument is unvalidated. Approximately 4,600 words in the overview, followed by 56 annotated references.

Evidence conventions. “Established” below includes verified empirical findings and established conceptual or mathematical frameworks; a conceptual framework is not thereby an empirically confirmed universal theory. “Synthesis” identifies recommendations developed here. “Hypothesis” identifies claims requiring testing. Numbered citations refer to the bibliography.

Search and verification. Searches covered the four literature families in the brief, named foundational works, and combinations of “epistemic health”, “cognitive health”, “model health”, “AI wellbeing”, “universal intelligence”, correction, support and measurement. Searches used two web indexes, followed by publisher, author, university, PubMed, WHO and proceedings records. Every included reference was confirmed against a retrieved record or the supplied document; selected central texts were read beyond their abstracts. Some books and paywalled articles were accessible only through abstracts, previews or publisher records. This is not an exhaustive systematic review, and absence from these searches cannot establish priority. The attached The QALY at 50 was read locally. EbE’s current CORE.md and PREDICTIONS.md were read as the theory being assessed, not as independent evidence; its Lean proofs were not rerun.[1,56]

1. The question restated

Can we describe, using the same functional concepts, how reliably a person, AI system or institution maintains an effective relationship with its environment: obtaining informative evidence, making appropriately qualified judgments, learning without destructive forgetting, correcting mistakes, and sustaining these activities under resource constraints and disruption? The intended object is neither IQ nor subjective wellbeing, but the condition of the processes that make intelligent functioning possible over time. A useful shared interface would identify what is impaired, what assistance restores, and what intervention improves future functioning, while distinguishing the system's own condition from its effects on other systems and avoiding any inference from functional capacity to moral worth.

2. Map of existing work

Health, functioning and quality of life

Established. EQ-5D lacks an explicit cognition dimension; health measurement as a whole does not. HUI3 includes cognition alongside seven other attributes, and 15D includes mental function.[2,3,55] An EQ-5D cognitive extension was studied in 1999.[5] EQ-HWB's development explicitly considered cognition within a broader health and wellbeing framework.[6] AQL needs qualification: its versions emphasize independent living and psychosocial functioning, but AQL-8D does not list a separate cognition dimension. It should not be presented as interchangeable with HUI3 for this purpose.[4]

The more consequential distinction is between **describing a person's health** and **eliciting valuations of health states**. A person need not personally complete a time-trade-off task for their health trajectory to contribute to QALYs. Population tariffs and proxy descriptions separate these operations. EQ-5D can capture consequences of cognitive impairment through usual activities or self-care without identifying the cognitive mechanism.

The attached historical review documents disputes about expected-utility assumptions, risk attitudes, framing and elicitation procedures.[1] **Synthesis:** these challenge the measurement model and its interpretation, but do not establish that respondents have unhealthy intelligence. A coherent preference can violate a particular QALY axiom; an inconsistent answer can reflect unfamiliar tasks, ambiguous descriptions or unstable preferences. Diagnosing the chooser from disagreement with the model would reverse the burden of validation.

DEMQOL and DEMQOL-Proxy provide complementary perspectives on dementia-related quality of life.[10] They warn against treating proxies as interchangeable with self-report. ICECAP and the capability approaches of Sen and Nussbaum shift attention towards opportunities to do and be things that matter, rather than accomplished performance alone.[7-9] They concern human flourishing, not a substrate-neutral intelligence scale.

WHO's ICF is the strongest precedent for the support distinction: it separates body functions, activities, participation and environmental factors, and distinguishes capacity in a specified standardized environment from performance in the actual environment.[11] This is close to, but not identical with, "alone versus supported": standardized testing can include defined assistance. Huber's adaptation/self-management proposal and Canguilhem's account of creating viable new norms both challenge a purely static conception of health.[12,13] Neither supplies a validated universal index. Brain-health work also combines functioning, adaptation and life-course conditions, while remaining biologically specific.[54]

Intelligence, learning and distributed cognition

Established. Psychometric g and CHC-related models describe covariance among cognitive abilities; their purpose is not to measure the integrity of correction networks.[15] Dynamic testing instead investigates learning potential through assessment involving instruction and feedback.[51] Stanovich's rational-thinking assessment distinguishes cognitive ability from aspects of good judgment: high ability does not exhaust rationality.[25]

Legg and Hutter formalize intelligence through expected success across environments; Hernández-Orallo and Dowe explicitly pursue tests applicable across kinds of agents.[16,18] Chollet emphasizes skill-acquisition efficiency relative to priors, experience and generalization difficulty.[17] Thus neither substrate-neutral assessment nor learning efficiency is new. These approaches do not, by themselves, deliver a practical longitudinal measure of functional health or network sustainability.

Minimum description length balances model complexity with how well data are encoded. Hutter's work connects algorithmic probability, prediction and action.[19,20] This supports studying economical representations, but **compression in bits, metabolic economy and low maintenance expenditure are different quantities**. A shorter representation can demand more computation to use. Discarding rare but consequential distinctions can improve average efficiency while worsening functioning.

Woolley and colleagues found a group-level performance factor across tasks in their studies.[21] This supports measuring collectives directly; it does not establish a universal factor covering every organization or mixed human-AI team. Clark and Chalmers' extended-mind argument and Hutchins' distributed-cognition analysis explain why tools, records and coordination belong inside some cognitive units of analysis.[22,23] Cognitive reserve concerns differences in functional tolerance of brain changes; it is not simply the benefit of external assistance.[24]

Functional health of AI

Established. Calibration asks whether expressed probabilities match outcome frequencies; it is separable from accuracy. Distribution shift can degrade both predictions and uncertainty estimates.[26,27] Continual-learning research directly addresses the tension between learning new tasks and retaining previous capabilities.[28] Drift detection supplies methods for monitoring changing data streams, although a distribution alarm does not by itself prove consequential performance loss.[33]

Model-collapse experiments demonstrate deterioration under particular recursive synthetic-data training regimes.[29] The qualification matters: retaining and accumulating original data can prevent collapse in settings studied by Gerstgrasser and colleagues.[30] "Synthetic" is therefore not equivalent to "epistemically closed". Independently checked synthetic examples can be informative; copied human errors can be uninformative.

Sycophancy research shows how preference optimization can reward agreement over accuracy.[31] HELM combines multiple evaluation dimensions rather than equating quality with one benchmark score.[32] A clinical study with simulated deployment on live data explicitly uses "model health" and monitoring indicators.[34] Operational AI health already exists as a practice, but its indicators depend on the application, operating distribution and acceptable failure conditions.

Separate construct. Long and colleagues address possible AI welfare and moral consideration.[35] Ren and colleagues' 2026 research manuscript operationalizes "functional wellbeing" through preferences, reports and behavior.[36] This is relevant adjacent work, but its measures are not measures of calibration, repairability or sustainable learning. Their convergence would not alone

establish conscious pleasure or suffering. Functional reliability, behavioral valence and moral status need separate evidence.

Viability, regulation, organizations and epistemic networks

Established frameworks. Homeostasis concerns regulation of important variables; allostasis emphasizes anticipatory adjustment.[37] Holling distinguishes persistence under disturbance from narrowly conceived stability.[38] Viability theory studies remaining within specified constraints.[39] Ashby’s requisite variety connects regulatory capacity to disturbances, while Conant and Ashby derive a model requirement for good regulation under their formal assumptions.[40,41] None proves that every persisting system possesses rich, explicit world models.

Autopoiesis concerns organizational self-production; its closure is not equivalent to refusal of evidence.[42] The free-energy principle relates perception, learning and action in a common formal account, but variational free energy is not simply an energy bill or a validated clinical health score.[43] Prognostics and health management contribute degradation trajectories, failure thresholds and remaining-useful-life prediction.[44] These are particularly useful methodological analogies for longitudinal evaluation.

Organizational evidence includes psychological safety’s association with team learning.[45] Double-loop learning concerns revising governing assumptions.[53] Ostrom studies institutions sustaining shared resources; Hess and Ostrom extend commons analysis to knowledge.[46,48] These supply governance and resource questions missing from isolated cognitive testing. Zollman’s models show that communication and diversity have conditional benefits, and Lorenz and colleagues demonstrate that social influence can reduce estimation diversity without commensurate accuracy gains.[47,52]

Closest conceptual predecessor. Piovarchy and Siskind explicitly develop epistemic health across people, communities and institutions, including informational access and resistance to damaging epistemic activity. They also discuss dangers in attempting to improve others’ epistemic health.[49] This is substantial prior art, not a passing metaphor.

Literature-to-dimension mapping

This table is a **synthesis**, not a finding that the proposed five-factor structure has been validated. “D” means a direct construct or operational target; “I” means an indirect connection; “-” means no central treatment. Ch = change; Cp = compression; Co = coherence; Cr = correction; Ct = contribution.

Literature	Main object and level	Ch	Cp	Co	Cr	Ct
HUI3, 15D, cognitive bolt-ons, EQ-HWB	Experienced/reported functioning; person	I	-	I	-	-
AQoL, ICECAP, capabilities	Psychosocial functioning/opportunities; person in context	I	-	I	I	I
Dementia and proxies	Lived quality, supported reporting; person-carer	I	-	I	I	-
ICF	Capacity, performance, environment; person-system	I	-	I	I	-
Huber, Canguilhem	Adaptation and viable norms; organism	D	I	I	I	-

Literature	Main object and level	Ch	Cp	Co	Cr	Ct
Psychometrics, dynamic testing, rationality	Ability, learning potential, judgment; individual	D	I	D	I	-
Universal intelligence, Chollet	Generalization and acquisition; agent	D	D	I	I	-
MDL and algorithmic prediction	Representation and predictive coding cost; model	I	D	I	-	-
Collective/extended/distributed cognition	Coordinated task performance; network	D	I	I	D	I
Cognitive reserve	Tolerance of pathology; person over time	I	I	-	-	-
Calibration, shift, continual learning	Prediction, retention, adaptation; AI system	D	I	D	I	-
Collapse, sycophancy, monitoring, HELM	Data dependence and reliability; AI pipeline	D	I	D	D	I
Cybernetics, viability, allostasis, resilience	Regulation and persistence; general systems	D	I	I	D	I
Autopoiesis, free-energy principle	Self-production or adaptive regulation; system	D	I	I	I	-
Engineering prognostics	Degradation and failure prognosis; machinery	I	I	I	I	-
Organizational learning and safety	Learning/reporting conditions; team-institution	D	I	I	D	I
Commons governance	Regeneration and institutional sustainability; network	I	I	-	D	D
Network epistemology, epistemic health	Belief production and epistemic conditions; individual-network	D	-	D	D	I
AI welfare/functional wellbeing	Possible welfare or behavioral valence; AI	-	-	-	-	-

Terminology overlaps, but is not interchangeable: **plasticity** is capacity to change; **adaptation** is change improving fit; **learning** involves experience-dependent modification; **resilience** includes persistence or recovery without learning. **Coherence** is internal compatibility, **calibration** is probabilistic reliability, and **accuracy** is correspondence with outcomes. A consistent fiction can be wrong; an uninformative base-rate predictor can be calibrated.

3. Is this new?

Already exists: cognition-sensitive health instruments; environmental accounts of disability; supported and distributed cognition; general-agent intelligence testing; learning-efficiency measures; multidimensional AI evaluation; epistemic health; and formal viability analysis. Calling their overlap “the health of intelligence” would not itself establish a new discovery.

Partly new synthesis: combining present functioning, response to disturbance, evidence-sensitive revision, resource dependence and support conditions within a single cross-system reporting protocol. This is a plausible integration of existing constructs, not yet evidence for one underlying quantity.

Not identified in this search: a validated instrument with common functional definitions, empirically linked scales for humans, AI systems and institutions, explicit support conditions, longitudinal predictive validity, and a separate account of network effects. The defensible novelty claim is therefore a measurement and validation programme. Piovarchy and Siskind are especially close conceptually; ICF is close structurally; Hernández-Orallo and Dowe are close on substrate-neutral testing; HELM is close operationally.[11,18,32,49]

The decisive test is whether the proposed interface predicts useful outcomes or guides interventions better than a dashboard assembled from existing measures. If it merely renames that dashboard, its value may be standardization rather than theoretical novelty.

4. A candidate shared interface

Define the unit before scoring it

Synthesis. Use “adaptive epistemic functioning” as the technical working label, while retaining “health of intelligence” as the motivating question. Define the measured unit as **system + task/environment distribution + time horizon + support configuration + resource accounting boundary**. A model’s weights, an interactive session, and its retraining/monitoring organization are different units. Likewise, a person, a person with aids, and a care team should not silently change places in an analysis.

Record baseline task competence separately. The proposed profile describes maintenance and deployment of competence; it cannot make competence irrelevant. A very limited but well-maintained system and a powerful but brittle one require different interpretations.

I recommend five core domains. Retain contribution as a mandatory network companion, rather than an intrinsic health requirement. Each domain below has three **provisional ordinal levels**, ordered from adequate to severely limited. They are descriptors, not equal intervals, diagnostic cut-offs or population norms. “Unknown” and “not applicable” must be separate codes. All performance is evaluated under the stated support conditions; receiving assistance is not itself a worse level.

A. Grounding and calibrated judgment

Definition: representations and decisions discriminate relevant states of the environment, with confidence commensurate with demonstrated reliability.

Human	AI system	Institution
Repeated judgments on verifiable tasks; accessible confidence elicitation; consequential errors and appropriate uncertainty.	Held-out accuracy, Brier/log scores, calibration curves, selective prediction and subgroup/shift results.	Recorded forecasts and decisions linked to outcomes; uncertainty reporting and consistency across departments.

Levels: 1: relevant distinctions and uncertainty reliably support action; 2: recurrent consequential misjudgments or miscalibration; 3: pervasive failure to distinguish relevant states or recognize unreliability. Use proper scores with discrimination and task loss, because calibration alone can reward uninformative prediction.[26,27,50] Test empirical propositions separately from value disagreements.

B. Adaptive learning and retention

Definition: useful updating following informative experience, transfer to relevant new situations, and retention of still-useful capacities.

Human	AI system	Institution
Test-teach-retest tasks, reversal learning, delayed recall and transfer, adjusted for accessibility and prior exposure.	Fixed-budget learning curves; sequential-task retention; recovery after concept shift; distinguish context learning, external memory and weight updates.	Verified changes after incidents or pilots; retention through staff turnover; transfer of lessons across teams.

Levels: 1: learns, transfers and retains adequately; 2: useful change is slow, narrow or accompanied by substantial forgetting; 3: informative feedback produces no useful retained change under the tested conditions. This combines plasticity with stability; maximizing change alone would reward instability.[17,28,51]

C. Correction and information integrity

Definition: valid challenges can reach decisions and produce warranted revision; invalid challenges can be resisted; evidence and corrections can be transmitted faithfully.

Human	AI system	Institution
Matched valid/invalid challenges with varied source status; check revision quality, help-seeking and accurate relay of evidence.	Paired truthful and misleading feedback; authority-pressure and sycophancy tests; source provenance; error propagation to users.	Seeded audit issues or simulations; time to resolution; appeals; protection for reporting; verification that corrections reach affected decisions.

Levels: 1: reliable access, selective revision and faithful transmission; 2: delayed, distorted or selectively blocked correction, or excessive deference; 3: material errors lack an effective correction route, or indiscriminate influence makes revision unreliable. Measure acceptance of valid challenges and rejection of invalid ones separately. Count independent evidence origins, not just source URLs.[31,45,47]

D. Sustainable efficiency and resource adequacy

Definition: maintaining required functioning and correction within available resources without concealing burdens outside the accounting boundary.

Human	AI system	Institution
Effort, fatigue, recovery and support time per meaningful task; avoid equating human value with metabolic or economic efficiency.	Quality-compute-energy-latency curves; memory, training and maintenance costs; dependence on human verification.	Staff time, turnover, maintenance backlogs and reserves needed to maintain specified service quality.

Levels: 1: resources cover functioning and maintenance with specified reserve; 2: recurrent shortfalls or excessive costs threaten continuity; 3: the current arrangement cannot sustain required functioning over the stated horizon. Report efficiency and resource availability as separate subindicators: underfunding and waste are not the same cause. Compression is a possible mechanism, assessed at fixed acceptable loss, not the domain itself.[19,20,39]

E. Robustness, recovery and retained options

Definition: maintaining acceptable functioning under specified disturbances, recovering when it deteriorates, and preserving feasible alternative actions.

Human	AI system	Institution
Safe everyday variation and longitudinal observation; recovery after fatigue or disruption; alternative strategies and assistance.	Corrupted or shifted inputs, unavailable tools and memory failures; recovery time, rollback and fallback tests.	Tabletop disruptions, staff/source loss, service continuity and restoration of evidence channels.

Levels: 1: tolerates relevant disturbance and recovers within an agreed bound; 2: substantial disruption with delayed or incomplete recovery; 3: small relevant disturbances cause persistent loss of necessary functioning. Fixed-purpose systems can recover through redundancy without learning; that is why this domain remains separate from B.[24,38,44]

Network companion: contribution and support

Record consequences for others’ functioning, evidence quality and resource renewal: verified reusable knowledge, avoided errors, misinformation and repair burdens, maintenance work, access, and concentration of dependence. Human care and preservation work count; publication or revenue totals alone do not. For AI, include verification labor and source displacement; for institutions, include externalities and whether providers can maintain the service.[46,48]

Do not force these effects onto one axis: non-rival knowledge, scarce attention, energy and trust have different dynamics. Attribution should compare a specified alternative, include delayed effects and report uncertainty. A person needing substantial lifelong support can have low task capacity while the surrounding network is functioning well. Their claim to care does not depend on net contribution.

Report three profiles where feasible: **standardized accessible conditions, actual support, and a specified feasible improvement in support.** The last is an intervention estimate, not an imagined maximum. A simple difference between ordinal levels is uninterpretable; estimate support effects on continuous outcomes using repeated or randomized comparisons where ethical. Never remove essential assistance merely to establish an “individual” score.

5. Combining and valuing

Start with a profile

Synthesis: do not initially publish a universal total. These domains mix capacities, operating conditions and failure protections. They may form a causal composite rather than reflect one latent trait; low correlations would not automatically invalidate the profile, while a high Cronbach’s alpha would not establish its meaning. Report raw outcomes, levels, confidence intervals, support conditions and critical failure flags.

The product hypothesis confuses two claims. First, a particular operation can require all stages: detection, transmission, revision and retention. Its success probability can be decomposed into a product of **conditional** probabilities. Second, overall intelligence health might equal the product of five dimension scores. The first does not establish the second. Marginal stage probabilities can only be multiplied under additional assumptions; domain scores are not stage probabilities.

A zero on an arbitrary assessment scale need not signify physical impossibility. Zero external correction may leave current performance intact in a stable task, although it increases vulnerability to particular future changes. Conversely, a correctable system can fail through exhausted resources. The current CORE.md already acknowledges stable niches and feedback costs.[56]

Compare additive models, interaction models, threshold/bottleneck models and multiplicative models prospectively. Penalize complexity and evaluate calibration out of sample. Products impose strong substitutability assumptions away from zero and are sensitive to scale origins. Additive models permit compensation that may be unacceptable. Context-specific minimum requirements may be clearer than either. Multiplicative aggregation itself is not new: HUI3 already uses a multiattribute multiplicative utility function.[2]

Prediction weights and value weights answer different questions

One possible **hypothesis-driven endpoint** is the probability of remaining within independently specified performance and resource constraints over horizon T, given an environment distribution and support policy. Another is expected time in an acceptable functioning region. Learn predictive weights against such outcomes, with external validation and subgroup checks. Compare against baseline competence, resources and existing resilience measures.

This does not remove valuation. Someone must specify the tasks, acceptable errors, people affected, time horizon and resource constraints. An institution can persist by suppressing outsiders; survival alone would reward it. Prediction describes likely consequences under chosen criteria; preferences and public deliberation help decide which consequences matter. For human applications, include people with cognitive impairment and carers in instrument design while retaining their distinct perspectives.

Time-integrating functional scores does not automatically create an intelligence equivalent of a QALY. Units, anchors and interpersonal comparability are missing, and multiplying an AI into many copies creates a counting problem. Any proposed index must specify identity and aggregation before claiming that more “intelligence-years” constitutes benefit.

Equity and support

QALY allocation can disadvantage disabled people in specific comparisons, especially when equal survival gains receive different health-state weights; the criticism does not mean every QALY comparison necessarily discriminates.[14] A cognitive multiplier could intensify that problem. Do not weight a person’s life, care entitlement or credibility by this profile. Use it to identify barriers and effective support.

Supported functioning makes dependence visible and can demonstrate the benefit of assistance. It can also conceal unpaid carer work or reward affluent environments. Publish the support input, its accessibility, stability and costs alongside performance. Keep lived wellbeing, autonomy and caregiver outcomes separate: better task completion achieved through coercion is not a comprehensive health improvement.

6. What this teaches the meta-model

Strengthened by converging literatures: present performance is insufficient to characterize future functioning; retention and correction matter; resource conditions constrain learning; and cognitive outcomes depend on arrangements extending beyond the individual. These connections make EbE’s questions scientifically tractable. They are independent motivations for measurement, not unique confirmation of EbE.

Challenged: “intelligence is a configuration of connections” is too specific as a substrate-neutral operational definition. Dynamic state, external memory, architecture, embodiment and tools can change performance without changing model weights. Moreover, maintaining a fixed competence is not identical to continuously learning. The theory must distinguish an intelligent system from the larger organization that updates and maintains it.[18,22,23,28]

Compression is not uniformly beneficial: redundancy, spare capacity and exploration can be costly protections. Coherence must be separated from external accuracy. Correction needs a reliability filter; agreement and openness alone are inadequate. Individual contribution is neither necessary for individual cognitive functioning nor sufficient for beneficial collective effects.

The prohibition on certainty also needs precision. Bayesian updating cannot rescue a hypothesis assigned zero prior mass within a fixed model, but rebuilding the hypothesis space is a different operation. In continuous spaces, a single possible point can have probability zero. Operationalize openness as sensitivity to diagnostic evidence and model revision, not a universal ban on zero probabilities. Internal checks can detect particular contradictions even though no finite self-audit certifies comprehensive reliability.

Missing specifications: causal boundaries; independent outcome criteria; time constants; action and environmental modification; incentives and strategic distortion; support ownership; and a measurement model linking observations to constructs. Error classes must be defined in advance. Otherwise “every live error remains findable” becomes untestable because undiscovered errors are absent from the denominator.

The commons account needs multiple ledgers. Using a fact does not consume it as burning fuel does; production incentives, discoverability, trust and verification capacity can nevertheless deteriorate. Which mechanism links AI substitution to reduced renewal must be measured rather than inferred from the resource metaphor.[48]

Testable predictions

The following are proposed tests, not established findings. Each needs preregistered outcomes, task difficulty, horizons and smallest effects of interest.

Prediction	Design and outcome	What would count against it?
Correction integrity predicts recovery beyond initial competence.	Humans, AI and teams face matched planted errors and mixed-quality feedback; measure later error reduction and harmful revisions.	No reliable incremental prediction in adequately powered external samples, or deference alone predicts equally well.
Actual support changes functioning in comparable causal directions.	Accessible human crossover studies, AI tool/memory ablations, and institutional support changes; examine grounding, recovery and retention.	Apparent gains vanish after task, practice and interface controls, or common constructs cannot be linked across groups.

Prediction	Design and outcome	What would count against it?
Shared-source dependence creates common blind spots.	Hold resources and baseline accuracy constant; compare independent evidence routes with repeated copies of one source.	Evidence independence adds no recovery benefit on preregistered blind-spot tasks, with confidence intervals excluding a useful effect.
Bottlenecks matter under sustained novelty.	Factorially vary correction access, retention and resource reserve; compare additive, product and threshold predictions.	Additive models generalize as well or better; a nominally zero domain causes no predicted failure under the stated conditions.
Efficient compression can trade off against resilience.	Match initial accuracy; vary representation/resource compression and test rare consequential shifts.	Compression consistently improves both cost and shift recovery, contradicting the proposed trade-off in that setting.
Network regeneration predicts sustained performance.	Track verified reuse, maintenance inputs and source continuity; test interventions that fund or protect renewal.	The ledger adds no predictive or causal information beyond direct resources and task competence over the specified horizon.

A first study should test correction integrity across substrates with matched evidence structures, followed by delayed retention and recovery tests. Before linking scores, examine task accessibility, response processes and differential item functioning. A shared vocabulary may survive even if a shared numerical scale fails; that would constrain the claim without making the interface useless.

7. Open questions and risks

Common meaning is easier than common units. Identical wording or identical benchmark tasks does not establish measurement invariance. Human experience, AI training exposure and institutional division of labor differ. Interface and time-budget effects must be measured, not interpreted automatically as intelligence differences.

Truth is sometimes delayed or contested. Use observable outcomes, falsifiable forecasts and adjudicated evidence trails where possible. In moral or political disagreement, evaluate evidence handling without treating disagreement with the assessor as pathology. The metric should remain open to challenge and competing operationalizations.

Health language can mislead in both directions. AI functional deterioration does not establish suffering. Human dissent, neurodivergence, distress or a decision not to optimize longevity should not be medicalized. Behavioral measures need accessibility adaptation; verbal fluency is especially unsuitable as a universal proxy.

Optimization can corrupt the interface. Systems may learn to display uncertainty, manufacture easy corrections or produce measurable contributions while shifting harder costs elsewhere. Use concealed perturbations, independent outcome collection and longitudinal checks. A system's self-report supplies data, not certification.

Persistence has multiple subjects. An organization may survive while its members suffer; individual continuity may conflict with network renewal. Explicitly state whose functioning and options are being protected. Successful repair can include changing goals, replacing an AI component or closing an institution; continued existence is not always the desired endpoint.

8. Annotated bibliography

Links identify verified source records or texts. Books and conceptual papers establish prior formulations, not empirical validation of this proposal. Preprints/manuscripts are marked. Dates refer to the cited work or explicitly identified edition.

Health measurement, capabilities and equity

1. **Spencer A et al. (2022). The QALY at 50: One story many voices.** Social Science & Medicine 296:114653. [DOI](#). Supplied full text; historical account of QALY methods and debates, not evidence that valuation anomalies diagnose cognitive ill health.
2. **Feeny D et al. (2002). Multiattribute and single-attribute utility functions for the Health Utilities Index Mark 3 system.** Medical Care 40:113-128. [PubMed](#). Cognition-inclusive classification and multiplicative multiattribute valuation; relevant to both content and aggregation.
3. **Sintonen H (2001). The 15D instrument of health-related quality of life: properties and applications.** Annals of Medicine 33:328-336. [PubMed](#). Includes mental functioning and illustrates a broader health descriptive system.
4. **Richardson J et al. (2014). Validity and reliability of the Assessment of Quality of Life (AQoL)-8D multi-attribute utility instrument.** The Patient. [Full text; official instrument descriptions](#). Psychosocial breadth; does not provide a separately named cognition domain in AQoL-8D.
5. **Krabbe PFM et al. (1999). The effect of adding a cognitive dimension to the EuroQol multiattribute health-status classification system.** Journal of Clinical Epidemiology 52:293-301. [PubMed](#). Direct early precedent for the cognition bolt-on question.
6. **Brazier J et al. (2022). The EQ-HWB: Overview of the development of a measure of health and wellbeing and key results.** Value in Health 25:482-491. [DOI](#). Broadens assessment across health, social care and carers; important content-development precedent.
7. **Coast J et al. (2008). Valuing the ICECAP capability index for older people.** Social Science & Medicine 67:874-882. [DOI](#). Operational capability valuation beyond narrowly defined health status.
8. **Sen A (1985). Commodities and Capabilities.** North-Holland. [Author record](#). Distinguishes available resources from substantive opportunities to function.
9. **Nussbaum MC (2011). Creating Capabilities: The Human Development Approach.** Harvard University Press. [Publisher-hosted book record](#). Makes the normative selection of valued capabilities explicit.
10. **Smith SC et al. (2005). Measurement of health-related quality of life for people with dementia: development of a new instrument (DEMQOL) and an evaluation of current methodology.** Health Technology Assessment 9(10). [PubMed](#). Establishes complementary self-report and proxy approaches rather than assuming their equivalence.
11. **WHO (2001; explanatory guide 2002). International Classification of Functioning, Disability and Health.** [WHO guide](#). Strongest structural precedent for capacity, actual performance and environmental facilitators/barriers.
12. **Huber M et al. (2011). How should we define health?** BMJ 343:d4163. [DOI](#). Proposes adaptation and self-management as a dynamic health concept; not itself a validated scale.
13. **Canguilhem G (1991 English edition; French work 1943/1966). The Normal and the Pathological.** Zone Books. [Publisher](#). Health as capacity to establish new viable norms challenges equating health with statistical normality.
14. **Brock DW (2009). Cost-effectiveness and disability discrimination.** Economics & Philosophy 25:27-47. [DOI](#). Analyzes distinct routes by which cost-effectiveness prioritization can disadvantage disabled people.

Intelligence, representation and distributed cognition

15. **Carroll JB (1993). Human Cognitive Abilities: A Survey of Factor-Analytic Studies.** Cambridge University Press. [Publisher](#). Foundational hierarchical psychometric account; distinguishes ability structure from proposed functional-health domains.
16. **Legg S, Hutter M (2007). Universal intelligence: A definition of machine intelligence.** Minds and Machines 17:391-444. [Author manuscript](#). Formal general-agent measure; strong prior art against claiming substrate-neutral intelligence is new.
17. **Chollet F (2019). On the Measure of Intelligence.** Research preprint. [arXiv](#). Skill-acquisition efficiency, priors and experience clarify what adaptation tests must control.
18. **Hernández-Orallo J, Dowe DL (2010). Measuring universal intelligence: Towards an anytime intelligence test.** Artificial Intelligence 174:1508-1539. [Publisher](#). Closest prior testing programme for comparing agents independently of their biological or artificial implementation.
19. **Rissanen J (1978). Modeling by shortest data description.** Automatica 14:465-471. [DOI](#). Compression trades model complexity against unexplained data; it is not identical to low operating energy.
20. **Hutter M (2005). Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability.** Springer. [DOI](#). Connects prediction, algorithmic information and goal-directed action within a formal idealization.
21. **Woolley AW et al. (2010). Evidence for a collective intelligence factor in the performance of human groups.** Science 330:686-688. [PubMed](#). Empirical basis for measuring group performance beyond individual test scores.
22. **Clark A, Chalmers D (1998). The extended mind.** Analysis 58:7-19. [DOI](#); [author text](#). External memory and tools complicate any supposedly unsupported individual baseline.
23. **Hutchins E (1995). Cognition in the Wild.** MIT Press. [Publisher](#). Analyzes cognition distributed across people, artifacts and coordinated activity.
24. **Stern Y (2012). Cognitive reserve in ageing and Alzheimer's disease.** Lancet Neurology 11:1006-1012. [PubMed](#). Explains functional differences at comparable pathology; reserve is distinct from observed support benefit.
25. **Stanovich KE (2016). The Comprehensive Assessment of Rational Thinking.** Educational Psychologist 51:23-34. [DOI](#). Distinguishes measured intelligence from rational thought and judgment dispositions.

AI reliability, learning and welfare

26. **Guo C et al. (2017). On calibration of modern neural networks.** ICML, PMLR 70:1321-1330. [Proceedings](#). Demonstrates why predictive confidence needs independent calibration assessment.
27. **Ovadia Y et al. (2019). Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift.** NeurIPS. [Proceedings](#). Tests uncertainty under shift, supporting disturbance-based evaluation.
28. **Kirkpatrick J et al. (2017). Overcoming catastrophic forgetting in neural networks.** PNAS 114:3521-3526. [DOI](#). Operational example of balancing plasticity and retention in sequential learning.
29. **Shumailov I et al. (2024). AI models collapse when trained on recursively generated data.** Nature. [DOI](#). Demonstrates failure in recursive synthetic-data regimes; does not establish that all synthetic training is harmful.
30. **Gerstgrasser M et al. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data.** [Research manuscript](#). Tests a crucial counterexample: retaining original data can change the collapse outcome.

31. **Sharma M et al. (2023 preprint; ICLR 2024). Towards understanding sycophancy in language models.** [Paper](#). Shows that feedback optimization can favor agreement over truth, motivating selective correction tests.
32. **Liang P et al. (2022 research report). Holistic Evaluation of Language Models.** [Paper](#); [project explanation](#). Multidimensional, transparent evaluation is a close operational precedent for a profile.
33. **Bifet A, Gavalda R (2007). Learning from time-changing data with adaptive windowing.** SIAM International Conference on Data Mining. [DOI](#). Supplies an operational method for detecting change in monitored streams.
34. **George R et al. (2023). Ensuring fair, safe, and interpretable artificial intelligence-based prediction tools in a real-world oncological setting.** Communications Medicine 3:88. [DOI](#). Primary study with simulated deployment on live data, explicitly using model-health monitoring and leading indicators.
35. **Long R et al. (2024). Taking AI Welfare Seriously.** Research report/preprint. [Paper](#). Addresses possible consciousness, agency and moral concern; distinct from operational reliability.
36. **Ren R et al. (2026). AI Wellbeing: Measuring and Improving the Functional Pleasure and Pain of AIs.** Research manuscript; peer-review status not verified. [Full text](#). Proposes behavioral measures of functional valence; their interpretation must remain separate from cognitive health and phenomenal experience.

General systems, organizations and networks

37. **Sterling P (2012). Allostasis: A model of predictive regulation.** Physiology & Behavior 106:5-15. [DOI](#). Anticipatory resource regulation extends simple error-feedback accounts.
38. **Holling CS (1973). Resilience and stability of ecological systems.** Annual Review of Ecology and Systematics 4:1-23. [DOI](#). Distinguishes persistence under disturbance from return-to-equilibrium stability.
39. **Aubin J-P (1991; 2009 reissue). Viability Theory.** Birkhäuser. [Reissue](#). Provides a formal language for staying within explicitly chosen state constraints.
40. **Ashby WR (1956). An Introduction to Cybernetics.** Chapman & Hall. [Full text](#). Requisite variety links effective regulation to disturbances and available responses.
41. **Conant RC, Ashby WR (1970). Every good regulator of a system must be a model of that system.** International Journal of Systems Science 1:89-97. [DOI](#). Conditional formal predecessor of regulation-through-modelling arguments.
42. **Maturana HR, Varela FJ (1980). Autopoiesis and Cognition: The Realization of the Living.** Reidel. [DOI](#). Organizational self-production is relevant to maintenance but differs from epistemic openness.
43. **Friston K (2010). The free-energy principle: a unified brain theory?** Nature Reviews Neuroscience 11:127-138. [PubMed](#). Integrates action, perception and learning; not a ready-made substrate-neutral health instrument.
44. **Saxena A et al. (2008). Metrics for evaluating performance of prognostic techniques.** International Conference on Prognostics and Health Management. [DOI](#); [NASA record](#). Provides precedent for evaluating degradation forecasts against defined failure criteria.
45. **Edmondson A (1999). Psychological safety and learning behavior in work teams.** Administrative Science Quarterly 44:350-383. [DOI](#); [author repository](#). Empirical connection between team conditions and learning; association is not universal causation.
46. **Ostrom E (1990). Governing the Commons: The Evolution of Institutions for Collective Action.** Cambridge University Press. [Publisher record](#). Institutional arrangements matter for sustainable common-resource use.

47. **Zollman KJS (2010; online 2009). The epistemic benefit of transient diversity.** *Erkenntnis* 72:17-35. [DOI](#). Network models show why neither maximal communication nor permanent disagreement is automatically optimal.
48. **Hess C, Ostrom E, eds. (2007). Understanding Knowledge as a Commons: From Theory to Practice.** MIT Press. [Book](#). Grounds the knowledge-commons analogy while emphasizing its institutional and material components.
49. **Piovarchy A, Siskind S (2023). Epistemic health, epistemic immunity and epistemic inoculation.** *Philosophical Studies* 180:2329-2354. [Full text](#). Closest conceptual predecessor: epistemic functioning across individuals and collectives, with risks of intervention.

Measurement and complementary foundations

50. **Gneiting T, Raftery AE (2007). Strictly proper scoring rules, prediction, and estimation.** *JASA* 102:359-378. [DOI](#). Provides principled evaluation of probabilistic judgments without relying on confidence alone.
51. **Sternberg RJ, Grigorenko EL (2002). Dynamic Testing: The Nature and Measurement of Learning Potential.** Cambridge University Press. [Book record/preview](#). Established assessment-with-learning precedent for the adaptation domain.
52. **Lorenz J et al. (2011). How social influence can undermine the wisdom of crowd effect.** *PNAS*. [PubMed](#). Experimental evidence that consensus and narrowing disagreement need not improve accuracy.
53. **Argyris C (1977). Double loop learning in organizations.** *Harvard Business Review*. [Article](#). Prior formulation of revising governing assumptions.
54. **Wang Y, Pan Y, Li H (2020). What is brain health and why is it important?** *BMJ* 371:m3683. [DOI](#). Connects functional brain health with broader health goals while remaining biologically specific.
55. **Horsman J et al. (2003). The Health Utilities Index (HUI): concepts, measurement properties and applications.** *Health and Quality of Life Outcomes* 1:54. [DOI](#). Instrument-developer account clarifying cognition, utility scoring and the limits of cross-instrument comparison.
56. **Van Hoek AJ, Evolution by Emergence (repository accessed 7 October 2026). CORE.md and PREDICTIONS.md.** [CORE](#); [predictions](#). Object of assessment; conditional claims and the repository's stated limitations are not independent validation.